
When does an optimal kernel become optimal? Finite-resolution kernel design for quantum simulation of dissipative dynamics by linear combinations of Hamiltonian simulations

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Abstract

The linear-combination-of-Hamiltonian-simulations (LCHS) method simulates the nonunitary propagator e^{-At} of a dissipative linear system as a weighted integral of Hamiltonian evolutions, and the choice of the weight kernel $\hat{f}(k)$ controls the precision–cost trade-off: exact kernels reach at best near-optimal precision scaling $\mathcal{O}(\log^{1/\beta} \varepsilon_{\text{LCHS}}^{-1})$, while recently derived analytic-extension (approximate) kernels reach the optimal $\mathcal{O}(\log \varepsilon_{\text{LCHS}}^{-1})$. We study, and then optimize, the kernel choice *at finite resolution* inside the most compact explicit LCHS circuit known to date — the Fejér–Clenshaw–Curtis (FCC) quadrature, single-QSP-selector implementation of Novikau and Joseph — under a constraint that is usually left implicit: the ℓ^1 norm of the discrete weights, which sets the circuit’s success probability, is held at parity with the established β -kernel baseline. Under this budget the standard asymptotic tuning of the Gaussian-regularized optimal kernels is misleading — it predicts they should lose to the β -family until $\varepsilon_{\text{LCHS}} \lesssim 10^{-6}$ — but a joint optimization of their regularization and contour-shift parameters reverses the verdict: at matched norm they need $1.5\text{--}2\times$ less truncation range than the β -family at every practically emulable precision, and additionally suppress a quadrature-aliasing failure mode of the sinusoidal grid that we identify. We then pose kernel selection as a small convex program on the fixed FCC grid — minimize the worst-case propagator error over the physically declared dissipation window subject to the norm-parity constraint — and show that the resulting optimized kernels restore the optimal $Q_A = \mathcal{O}(\log \varepsilon_{\text{LCHS}}^{-1})$ query scaling *with practical constants*: across end-to-end, gate-level emulations of the advection–diffusion benchmark, the measured marginal query cost of precision drops threefold — from $dk_{\text{max}}/d \log \varepsilon_{\text{LCHS}}^{-1} \approx 6$ and rising (β -family) to $1.8\text{--}2.0$, stable, at the norm-budget bound $2/\ln(2\alpha_0) = 2.09$ — at matched accuracy and matched success probability, with $2.5\text{--}3.5\times$ fewer block-encoding queries and truncation ranges across the emulable window and one fewer Fourier qubit at most accuracy levels. (We also show why the asymptotic exponent itself is not independently measurable over the practically emulable precision window — a caution for benchmarking optimal-scaling claims generally.) All weight constructions — including the kernel-dependent QSVT weight subroutine, whose phase factors we re-derive with an exact circuit-convention map and verify to 10^{-14} — are executed on a digital emulator of fault-tolerant hardware, and every kernel family is mapped to the $(n_k, \varepsilon_{\text{LCHS}})$ region where it wins. A spectral-interpolation diagnostic shows the remaining error at small n_k is informational rather than fundamental, delineating how far kernel design can go.

1 Introduction

Quantum computers implement unitary dynamics natively, but the equations that matter across plasma physics, fluid dynamics, finance, and chemistry are predominantly dissipative: $\partial_t \psi = -A\psi$ with a non-Hermitian generator A whose propagator e^{-At} contracts. Among the strategies for embedding such dynamics in unitary circuits — dilation to linear systems [13–15], time marching [16, 17], and Schrödingerization [10–12] — the linear combination of Hamiltonian simulations (LCHS) [1, 2] has emerged as particularly attractive because of its state-preparation optimality and high success probability. LCHS writes

$$e^{-At} = \int_{-\infty}^{\infty} \frac{\xi(k)}{1 - ik} e^{-i(A_H + kA_L)t} dk, \quad A = A_L + iA_H, \quad A_L \succeq 0, \quad (1)$$

so that a single auxiliary Fourier register and a family of *Hermitian* generators $C_k = A_H + kA_L$ replace the nonunitary flow. The kernel ξ controls everything quantitative. The original Cauchy kernel $\xi_{\text{sp}} = 1/[\pi(1 + ik)]$ requires a truncation range $k_{\text{max}} = \mathcal{O}(\varepsilon_{\text{LCHS}}^{-1})$ [1]; the exact kernel family of An, Childs, and Lin [2],

$$\xi_{\beta}(k) = (2\pi e^{-2^{\beta}} \exp[(1 + ik)^{\beta}])^{-1}, \quad \beta \in (0, 1), \quad (2)$$

improves this to the near-optimal $k_{\text{max}} = \mathcal{O}(\log^{1/\beta} \varepsilon_{\text{LCHS}}^{-1})$, with a no-go theorem forbidding exact kernels from doing better; and the recent approximate kernels based on analytic extensions of the decaying exponential [3–5] achieve the optimal $k_{\text{max}} = \mathcal{O}(\log \varepsilon_{\text{LCHS}}^{-1})$.

Which of these mathematical improvements survives contact with an actual circuit at the resolutions a real device (or emulator) can afford? This question separates cleanly into two ingredients that are often conflated. The *implementation* — how the integral (1) is discretized and compiled into selector, block-encoding, and weight-preparation circuits — was addressed by Novikau and Joseph [7], whose sinusoidal-coordinate construction $k = k_{\text{max}} \sin \theta$ yields Fejér–Clenshaw–Curtis (FCC) quadrature classically and, quantumly, lets a *single* QSP circuit with $\mathcal{O}(1)$ ancillae perform an exponential number of Hamiltonian simulations in parallel; it is, to our knowledge, the most compact explicit LCHS circuit in the literature. The *mathematical kernel* inside that implementation, however, was inherited from the near-optimal family (2), and the optimal-kernel constructions have so far been paired only with a more expensive multiplexed selector requiring $2\text{--}4\times$ larger ancilla registers [3]. The two best halves of the algorithm have not been combined, and it has been an open question — explicitly posed in [7] — whether the optimal kernels help at practical resolution.

This paper answers that question quantitatively, and the answer has an unexpected structure. Our contributions:

1. Norm-parity as the correct comparison rule. The ℓ^1 norm $\alpha = \sum_j |w_j|$ of the discrete LCHS weights fixes the success amplitude of the linear-combination-of-unitaries (LCU) sandwich; comparing kernels at different α silently trades success probability for truncation range. The analytic-extension kernels of [3] carry an explicit norm inflation e^c tied to their contour-shift parameter c ; imposing parity with the $\beta = 0.7$ baseline ($\alpha_0 = 1.3050$) forces $c \leq \ln \alpha_0 \approx 0.266$, and the family’s standard asymptotic parametrization [$\gamma = \sqrt{c + \log(M/\varepsilon_{\text{LCHS}})}/c$, $R = 2c\gamma^2$, giving $R \approx 2 + (2/c) \log(M/\varepsilon_{\text{LCHS}})$] then predicts a truncation range *worse* than the β -family’s until $\varepsilon_{\text{LCHS}} \sim 10^{-6}$. That prediction is an artifact of the parametrization: jointly optimizing (γ, c) against the exact tail integrals and transition errors shows the parity-constrained family in fact needs 25–37% *less* range than the β -kernel on worst-case bounds (k_{max} ratio 0.75 at $\varepsilon_{\text{LCHS}} = 10^{-2}$ falling to 0.63 at 10^{-8}), and $1.5\text{--}2\times$ less on the exact instance error at matched $(n_k, \varepsilon_{\text{LCHS}})$. Getting this comparison right requires computing with the kernels, not with their asymptotic envelopes — a recurring theme of this paper.

2. A quadrature-aliasing mechanism that reverses the verdict at fixed n_k . At fixed Fourier-register size n_k the dominant error is not truncation but an aliasing resonance of the FCC grid: the near-uniform central portion of the sinusoidal grid produces a spurious image of the propagator at dissipation depth $x^* \simeq 2(N_k - 1)/k_{\text{max}}$, which strikes whichever spectral mode lands near it. The Gaussian regularizer of the optimal kernels suppresses exactly the weight content that feeds this image, so at fixed $(n_x, n_k) = (6, 6)$ the parity-constrained Gaussian-regularized kernel reaches a measured $\varepsilon_{\text{LCHS}} = 1.1 \times 10^{-2}$ where the β -family saturates at 3.3×10^{-2} — a genuine, implementation-level advantage invisible to truncation-only analyses.

3. Finite-resolution kernel optimization. Once the grid, the selector, and the norm budget are fixed, the discrete kernel is just a vector $w \in \mathbb{C}^{N_k}$ entering the propagator error *linearly*. We therefore pose kernel choice as a convex program: minimize the worst-case propagator error over the declared dissipation window $x \in [0, \|A_L\|t]$, weighted by a declared problem-class envelope, subject to $\sum_j |w_j| \leq \alpha_0$ and Hermitian symmetry. The solution — an “optimized kernel” computable in under a second by linear programming — inherits $\mathcal{O}(\log \varepsilon_{\text{LCHS}}^{-1})$ scaling by a feasibility sandwich (the parity-constrained analytic-extension kernels are feasible points) and dominates every closed-form family at every resolution we tested, reaching $\varepsilon_{\text{LCHS}} = 10^{-4}$ -class accuracy with $k_{\text{max}} = 16$ where the baseline requires $k_{\text{max}} \gtrsim 40$ and one additional Fourier qubit.

4. End-to-end, gate-level validation with a reworked weight subroutine. Kernel substitution is not a paper-only change: the reference implementation hard-codes the β -family weights inside its weight-preparation oracle. We add a kernel-agnostic weight-selection option to the LCHS code (verified bit-identical against the native path), and rework the QSVT weight construction end to end: Chebyshev interpolation of $\sqrt{w}$ at the FCC nodes (which are Chebyshev–Gauss–Lobatto points in $\sin \theta$), even/odd phase-factor solves against an exact replica of the code’s QSVT convention — including a convention map we derive, $R(x) = -i e^{i\pi Z/4} W_x e^{i\pi Z/4}$, connecting the implementation’s reflection-form iterate to textbook W_x signal processing — with phase residuals at machine precision and emulator-verified weight states accurate to 10^{-13} . All headline claims are measured on the QuCF digital emulator [9] through the full circuit (initialization with amplitude amplification, weight oracles, selector, block-encodings), not from analytic estimates.

Combined, these give the first demonstration that optimal-scaling LCHS kernels can live inside the minimal known LCHS circuit — and a sharpened understanding of when they should: the marginal query cost of precision drops threefold, to the value the success-probability budget provably permits (Sec. 7), while the constant-factor gains at $\varepsilon_{\text{LCHS}} \in [10^{-3}, 10^{-4}]$ come mostly from the finite-resolution optimization rather than from asymptotic kernel theory — whose exponent distinction we show is not even independently measurable in this window.

2 LCHS in a minimal implementation

We work entirely inside the implementation framework of [7], held fixed throughout. The integral (1) is discretized by $k_j = k_{\text{max}} \sin \theta_j$, $\theta_j = -\pi/2 + j\Delta\theta$, $\Delta\theta = \pi/(N_k - 1)$, $j = 0, \dots, N_k - 1$, $N_k = 2^{n_k}$:

$$U_{\text{LCHS}} = \sum_{j=0}^{N_k-1} w_j e^{-iC_j t}, \quad C_j = A_H + \sin(\theta_j) k_{\text{max}} A_L, \quad w_j = k_{\text{max}} \cos(\theta_j) \Delta\theta W(k_j), \quad (3)$$

with the *weight density* $W(k) := \xi(k)/(1 - ik)$. Classically (3) is FCC quadrature at Chebyshev extrema; quantumly the entire sum is realized by one LCU sandwich $(O_{\sqrt{w}}^{\text{AA}}[U_{\sqrt{w}}^\dagger])^\dagger \circ S \circ O_{\sqrt{w}}^{\text{AA}}[U_{\sqrt{w}}]$, where the selector S is a single QSP sequence over the qubitization iterate of a block-encoded C_j whose k -dependence enters through an n_k -controlled-rotation $\sin \theta$ circuit, and the weight oracle prepares $\sum_j \sqrt{w_j} |j\rangle$. The block-encoding normalization of the reference build is $\Lambda(k_{\text{max}}) = 777.8125 k_{\text{max}}$ per unit time for the benchmark generator of Sec. 6 (this rule reproduces the reference build’s committed block-encoding angles to seven digits on its A_H branch, and the encoded operator extracted from the emulator’s matrix dump equals C_j/Λ to 9×10^{-17}), so the selector’s QSP evolution phase is $\tau = \Lambda(k_{\text{max}}) t$ and its cost is strictly proportional to k_{max} at fixed t : every kernel is charged the same price per unit of truncation range.

Query and gate accounting. We define the headline query count Q_A as the number of controlled applications of the block-encoding oracle (forward or inverse) inside the selector; per QSP angle pair the circuit applies the qubitization step W and its inverse, each containing one controlled block-encoding and one controlled inverse, so $Q_A = 4 \cdot \frac{N_{\text{ang}} - 1}{2} \cdot n_{\text{rep}}$ with $N_{\text{ang}} = 2761$ for the $\tau_0 = 1244.5$ angle table used by every run and $n_{\text{rep}} = \Lambda(k_{\text{max}})t/\tau_0 \in \mathbb{N}$. Gate counts are the emulator’s native single-target multi-controlled (STMC) totals for the full circuit — initialization, both weight oracles with their amplitude amplification, selector, and all block-encoding internals — so no cost is externalized to an abstract oracle. Success probability is $p = \sum_x |\langle x, 0_{\text{anc}} | \Psi \rangle|^2$, the squared norm of the zero-ancilla block of the final state. The end-to-end error is $\varepsilon_{\text{LCHS}} = \|\hat{u} - \hat{v}\|_2$

where $\hat{u}$ is the normalized, phase-aligned zero-ancilla output on the state register and $\hat{v} \propto e^{-At}\psi_0$ is computed at runtime from the same discrete generator by exact diagonalization.

Error anatomy. For the benchmark below A is circulant, hence normal, and every claim about the discretized kernel can be checked exactly: per spatial Fourier mode q with eigenvalue $\lambda_q = \ell_q + i h_q$,

$$\left| e^{-\lambda_q t} - \sum_j w_j e^{-i(h_q + k_j \ell_q)t} \right| = \left| e^{-x_q} - S_N(x_q) \right|, \quad S_N(x) := \sum_j w_j e^{-i k_j x}, \quad x_q = \ell_q t, \quad (4)$$

so the whole discretization problem collapses to how well the exponential sum S_N tracks e^{-x} on the *dissipation window* $x \in [0, \|A_L\|t]$. This scalar object is the design surface for everything that follows; the emulator then verifies that the compiled circuit realizes it. (Our classical design evaluations of (4) predict the emulator’s end-to-end error to $\sim 1\%$: e.g. 2.106×10^{-2} predicted vs. 2.087×10^{-2} measured at $(n_k, k_{\max}) = (7, 20)$, $t = 0.8$.)

3 Kernel families under a success-probability budget

Exact kernels. The Cauchy kernel gives real positive weights with minimal norm $\alpha = 1$ but $1/k^2$ decay. The β -family (2) satisfies the four admissibility conditions of exact LCHS (decay, normalization $\int W dk = 1$, continuity, lower-half-plane analyticity) and decays as $\exp[-\cos(\beta\pi/2)|k|^\beta]$; its measured ℓ^1 norm at $\beta = 0.7$ is $\alpha_0 = 1.3050$, and we adopt this number as the *norm-parity budget* for every kernel in this paper. For the benchmark, $\beta \in [0.7, 0.8]$ is the empirically near-optimal range [2, 7], which our landscape scans confirm.

Approximate (analytic-extension) kernels. Low and Somma [3] construct kernels whose scalar transform approximates e^{-x} on $x \geq 0$ rather than reproducing it exactly, evading the no-go theorem of [2]. Their two-sided kernel, in the convention of (1)–(3), is the Gaussian-regularized, contour-shifted Cauchy density

$$W_{f_2}(k; \gamma, c) = \frac{1}{\pi} \frac{e^{c(1-ik)} e^{-(k^2+1)/(4\gamma^2)}}{1 + k^2}, \quad (5)$$

$$G(x) = \frac{1}{2} e^{-x} \operatorname{erfc}\left(\frac{1}{2\gamma} - \gamma(x+c)\right) + \frac{1}{2} e^{x+2c} \operatorname{erfc}\left(\frac{1}{2\gamma} + \gamma(x+c)\right),$$

where $G(x) = \int W_{f_2} e^{-ikx} dk$ obeys $|G(x) - e^{-x}| \leq \mathcal{O}(e^{2c} \operatorname{erfc}(\gamma c))$ for $x \geq 0$ (we verified (5) numerically to 10^{-16}). Truncation costs $k_{\max} \geq R \approx 2c\gamma^2$ with $\gamma = \sqrt{c + \log(M/\varepsilon_{\text{LCHS}})}/c$, i.e. $R \approx 2 + (2/c) \log(M/\varepsilon_{\text{LCHS}})$, while the ℓ^1 norm is $\alpha_{f_2} \leq e^c$: *the optimal scaling is purchased with success probability*. Two remarks matter in practice. First, the shift phase e^{-ikc} can be absorbed exactly into the generator, $A_L \rightarrow A_L + (c/t)\mathbb{H}$, making the weight function real and even and halving the QSVT weight-construction cost — but the recovered norm reappears as e^{-c} signal decay, so the success-probability price is representation-independent; we therefore keep the phase in the weights, which leaves the block-encoding untouched. Second, the family must be tuned jointly, not asymptotically. Under the parity budget one must take $c \leq \ln \alpha_0 \approx 0.266$; the asymptotic coupling $\gamma = \sqrt{c + \log(M/\varepsilon_{\text{LCHS}})}/c$ then gives $R(\varepsilon_{\text{LCHS}}) \approx 2 + 7.5 \log(M/\varepsilon_{\text{LCHS}})$, nominally worse than the $\beta = 0.7$ range until $\varepsilon_{\text{LCHS}} \sim 10^{-6}$. Optimizing (γ, c) jointly against the exact truncation and transition errors instead yields ranges 25–37% *below* the β -kernel’s worst-case requirement across $\varepsilon_{\text{LCHS}} \in [10^{-2}, 10^{-8}]$, and our exact per-mode design scans confirm the advantage on the instance error at matched $(n_k, \varepsilon_{\text{LCHS}})$ (e.g. $k_{\max} = 16$ vs. 28 at $n_k = 9$, $\varepsilon_{\text{LCHS}} = 10^{-3}$). The one-sided (f_1) and rational ($f_{j,y}$) members shift constants but not conclusions; the Fourier-transform kernel constructions of [6] likewise improve universal constants, quoted at $\varepsilon_{\text{LCHS}} \leq 10^{-8}$.

4 Finite-resolution kernel optimization

At fixed grid the map $w \mapsto S_N$ in (4) is linear, and the FCC pairing $k_{N-1-j} = -k_j$ makes S_N real for Hermitian-symmetric weights ($w_{N-1-j} = \bar{w}_j$). We therefore define the *optimized kernel* (used as the primary substituted kernel below) as the solution of the linear program

$$\min_{w \in \mathbb{C}^{N_k}} \max_{x \in X} \rho(x)^{-1} |S_N(x) - e^{-x}| \quad \text{s.t.} \quad \sum_j |w_j| \leq \alpha_0, \quad w_{N-1-j} = \bar{w}_j, \quad (6)$$

where X is a dense grid on the full dissipation window $[0, \|A_L\|t]$ and $\rho \geq 1$ is a declared problem-class envelope ($\rho(x) = \min\{\rho_{\text{cap}}, 1/E(x)\}$ with E the spectral envelope of Gaussian initial data of width $\geq \sigma_c$ mapped through the generator’s dispersion relation; $\sigma_c = 0.05$, $\rho_{\text{cap}} = 3 \times 10^3$). The program uses only information the implementation already exploits — the operator symbol and the class of initial data, the same inputs that fix k_{max} and k_{min} in [7] — and none of the instance’s spectrum or initial condition; the same weights transfer unchanged across spatial resolutions (the $(n_k, k_{\text{max}}) = (6, 4)$ design evaluates, exactly, to 1.03×10^{-2} on the $n_x = 6$ instance — emulator-confirmed — and 9.4×10^{-3} on the finer $n_x = 7$ instance). With the modulus handled by a 16-gon outer approximation (tightened by $\cos(\pi/16)$ so the true norm respects the budget), (6) is a linear program solved by HiGHS in 0.2 s ($n_k \leq 7$) to 90 s ($n_k = 10$) per configuration.

Proposition 1 (scaling sandwich). *For every (γ, c) with $\alpha_{f_2}(\gamma, c) \leq \alpha_0$, the discrete f_2 weights are feasible for (6); hence the optimized kernel’s truncation range obeys $k_{\text{max}}^{\text{opt}}(\varepsilon_{\text{LCHS}}) \leq 2 + (2/c^*) \log(M/\varepsilon_{\text{LCHS}}) = \mathcal{O}(\log \varepsilon_{\text{LCHS}}^{-1})$ with $c^* = \ln \alpha_0$, while $k_{\text{max}}^{\text{opt}}(\varepsilon_{\text{LCHS}}) = \Omega(\log \varepsilon_{\text{LCHS}}^{-1})$ by the precision lower bound for linear-ODE solvers. The optimized kernel is therefore optimal-scaling, with instance constants no worse than every admissible closed-form family.*

An informational ceiling. How much of the residual error at small n_k is fundamental? For a normal generator the answer is: none of it. The instance’s dissipation spectrum $\{x_q\}$ contains at most $N_x/2 + 1$ distinct values; a weighted least-squares solve of $S_N(x_q) = e^{-x_q}$ (i.e., dropping the window in (6) in favor of the exact spectral points) drives the discretization error to machine precision ($\sim 10^{-11}$) already at $n_k = 5$, $k_{\text{max}} = 4$, with ℓ^1 norms 1.25–1.7. We report this *spectral-interpolation diagnostic* for what it is — a limit case that presumes exact spectral knowledge and has no truncation-theoretic content — because it cleanly separates the two readings of the small- n_k error wall: the wall is *informational* (what the kernel is allowed to know), not *fundamental* (what N_k degrees of freedom can express).

5 Reworking the weight construction

The reference build hard-codes the β -family weights inside its weight-preparation oracle (a multiplexed rotation cascade computing $\phi_{y,j} = 2 \arccos |\sqrt{w_j}|$, $\phi_{z,j} = -2 \arg \sqrt{w_j}$ from (2) in source). Substituting a kernel therefore requires reworking this subroutine, which we do twice over:

(a) *Weight-table option.* We add a kernel-agnostic branch to the weight oracle that reads the per-node angle pair (ϕ_y, ϕ_z) from a file computed and verified externally for any kernel. Fed the β -family angles, the new path reproduces the native build’s full LCHS output bit-for-bit (maximum amplitude difference 6.7×10^{-16}), so baseline and substituted kernels run through interchangeable machinery.

(b) *QSVT path.* For the QSVT construction, $\sqrt{w}$ is interpolated at the nodes $s_j = \sin \theta_j$ — which are exactly Chebyshev–Gauss–Lobatto points — by an even real polynomial $P_e(s) \approx \Re \sqrt{w}/\nu$ and an odd one $P_o(s) \approx \Im \sqrt{w}/\nu$ (interpolation residual $\leq 1.1 \times 10^{-15}$), each realized by a QSVT sequence over the one-ancilla $\sin \theta$ block-encoding and combined by a one-qubit LCU. The phase factors are computed by our own solver against an *exact replica* of the code’s QSVT circuit. Two convention facts, derived and then verified on the replica, make this work: the code’s reflection-form iterate obeys $R(x) = -i e^{i\pi Z/4} W_x(x) e^{i\pi Z/4}$, mapping its phases to textbook W_x -convention symmetric-QSP phases shifted by $\pi/4$ at the ends and $\pi/2$ in the middle; and the code’s even-length (odd-degree) sequences realize $i \sigma P_o(x)$ — the imaginary unit needed by the complex-weight LCU appears natively. Phases are then obtained by homotopy-continued Levenberg–Marquardt on the symmetric ansatz of [21] with residuals $\leq 5 \times 10^{-16}$, and the standalone weight oracle measured on the emulator reproduces the target $\sqrt{w_j}/(2\nu\sqrt{N_k})$ amplitudes to 4.3×10^{-14} (β -family) and 1.9×10^{-13} (optimized kernel). One practical caution we can document from experience: the inverse-side oracle of the LCU sandwich must prepare the *conjugate* weights (odd polynomial negated); reusing the forward phases silently collapses the sandwich onto $\sum_j |w_j|$ — discarding every kernel phase — with a measured error of 1.0×10^{-1} at the baseline headline configuration before we caught it. Full LCHS runs through the QSVT weight path agree with the weight-table path at matched configuration (Sec. 7); the scaling sweeps use the weight-table path, whose ancilla footprint matches the reference build’s.

6 Experimental setup

The benchmark is the reference paper’s advection–diffusion problem: periodic $x \in [0, 1)$, $\partial_t \psi = -v \partial_x \psi + D \partial_x^2 \psi$ with $v = 1$, $D = 0.01$, central differences on $N_x = 2^{n_x}$ points ($\Delta x = 1/(N_x - 1)$), Gaussian initial data ($x_c = 0.5$, $w_d = 0.05$) prepared by the reference initialization circuit with two rounds of amplitude amplification; primary time $t = 0.8$, with transfer checks at $t \in \{0.4, 1.6\}$. The circulant generator gives $\|A_L\|t \approx 127$ at $t = 0.8$. All runs use the QuCF emulator (double precision, GPU) with the reference selector-angle table ($\tau_0 = 1244.5$, QSP tolerance 10^{-16}) and this paper’s deck generator, whose block-encoding angles reproduce the reference build’s on their common configurations. As an anchor, our rebuilt emulator reproduces the authors’ committed reference output at $(n_k, k_{\max}) = (7, 20)$ *bit-identically* (same 2,692,135 STMC gates, same $p = 0.21507$, zero amplitude difference). Baseline kernels use the paper’s $\beta = 0.7$ (its near-optimal choice, also best in our scans); characterization adds $\beta \in \{0.5, 0.8, 0.9\}$, the Cauchy kernel, the parity-tuned f_2 family, the optimized kernel, and the spectral-interpolation diagnostic. The precision-scaling sweep targets $\varepsilon_{\text{LCHS}} \in \{10^{-1.5}, 10^{-2}, 10^{-2.5}, 10^{-3}, 10^{-3.5}, 10^{-4}\}$ along each kernel’s minimal-resource frontier (n_k minimal, then $k_{\max}$ minimal), plus the fixed-(6, 6) restriction; the design grid, all hyperparameters, and the committed sweep were fixed before the validation runs (plan and logs in the supplementary code).

Three implementation details are disclosed for full transparency. (i) Selector angles are composed by integer repetition of the reference τ_0 table rather than regenerated per configuration; this costs $\sim 4\%$ extra QSP overhead per unit time, identically for every kernel, and makes Q_A exactly proportional to the truncation range. (ii) Our deck generator resolves a $\sim 5 \times 10^{-4}$ internal normalization inconsistency between the two block-encoding branches of the reference input decks in favor of the branch that sets the QSP clock; the resulting circuits agree with the reference decks to 10^{-4} in final amplitudes on their common configurations, and identically-generated decks are used for baseline and variants alike. (iii) Kernel hyperparameters (β ; γ , c ; the LP weights) were selected by the pre-committed classical design scans; no hyperparameter was revised after the emulator validation runs. (iv) One sweep entry (the $t=1.6$ baseline) was initially instantiated at a guessed $k_{\max} = 8$ rather than its design-scan optimum $k_{\max} = 4$; the scan-optimal configuration was added when the omission was noticed, and both runs are reported (the guessed one turns out to sit on an aliasing resonance and appears in the failure analysis).

7 Results

Headline comparison at fixed quantum memory. At the memory-constrained configuration $(n_x, n_k) = (6, 6)$, $t = 0.8$, the β -family baseline saturates: its best achievable accuracy over all truncation ranges is $\varepsilon_{\text{LCHS}} = 3.3 \times 10^{-2}$ (at $k_{\max} = 10$, $Q_A = 27,600$ selector queries, $p = 0.238$) — consistent with the reference paper’s observation that per-mille accuracy required $n_k \geq 9$. The optimized kernel at $k_{\max} = 4$ reaches $\varepsilon_{\text{LCHS}} = 1.0 \times 10^{-2}$ — $3.3\times$ *more accurate* — with $Q_A = 11,040$ queries ($2.5\times$ fewer), $2.5\times$ fewer STMC gates, the same ancilla count, and success probability $p = 0.215$. On success probability one comparison subtlety deserves care: the baseline’s raw $p = 0.238$ at this configuration exceeds its own converged value precisely *because* its 3.3×10^{-2} error inflates the output norm with spurious amplitude; at every accuracy-converged configuration in Table 1 both kernels sit at $p = 0.213$ – 0.216 , and no run of the substituted kernel falls below the reference build’s converged level $p \approx 0.215$. Even the $k_{\max} = 2$ design (a single-repetition selector, $Q_A = 5,520$) reaches $\varepsilon_{\text{LCHS}} = 5.6 \times 10^{-2}$ — within a factor 1.7 of the baseline’s saturation accuracy at one fifth of its queries. The parity-tuned analytic-extension kernel $\hat{f}_2(\gamma=4, c=0.26)$ lands between the two, as the design theory predicts (Table 1).

Precision scaling. Along the minimal-resource frontier (Table 1, lower block; Fig. 1 left), the baseline requires $(n_k, k_{\max})$ growing from (7, 10) at $\varepsilon_{\text{LCHS}} \approx 10^{-1.5}$ to (11, 40) at the 10^{-4} class, while the optimized kernel spans the same accuracy range within (6, 4) to (10, 16) — 2.5 – $3.5\times$ fewer selector queries at every matched target, with the ratio growing toward deeper precision. How should the scaling itself be quantified? An honest answer requires admitting what a 2.3-decade window can and cannot measure. Because the kernels’ multiplicative constants sit *inside* the logarithm, both frontiers are locally power-like with slowly varying effective exponents, and free-exponent fits of $Q_A = a[\log(1/\varepsilon_{\text{LCHS}}) - L_0]^\gamma$ — to the emulator points or to the continuous classical design frontiers (which the emulator meets or slightly beats at every point; measured-to-design ratio 0.71–1.00, the

sub-unity values reflecting the phase-aligned normalized-state metric of the measurement) — return overlapping estimates, $\gamma \in 1.3\text{--}2.6$ across model variants with individual uncertainties of 0.6–0.8 (all variants tabulated in the logs): *the asymptotic exponent distinction between $\Theta(\log^{1.43})$ and $\Theta(\log)$ is not independently measurable in the practically emulable window.* What is measurable, sharply, is the *marginal query cost of precision*, $dk_{\max}/dL$ with $L = \log(1/\varepsilon_{\text{LCHS}})$ — how many extra truncation units each additional e -fold of accuracy costs:

$$\left(\frac{dk_{\max}}{dL}\right)_{\beta} = 5.87 \pm 0.21 \text{ (rising to 6.08 on the deeper half)}, \quad \left(\frac{dk_{\max}}{dL}\right)_{\text{opt}} = 1.84 \pm 0.04, \quad (7)$$

a $\sim 3\times$, many-standard-error separation, with the emulator-point fits giving the same picture (5.90 ± 0.73 vs. 1.93 ± 0.15). The optimized kernel’s slope sits at, and slightly below, the norm-budget constant $2/\ln(2\alpha_0) = 2.09$ derived in Sec. 4 (the declared class envelope buys the small residual) — direct evidence that it operates on the optimal $\Theta(\log \varepsilon_{\text{LCHS}}^{-1})$ law at the constant its success-probability budget permits — while the β -family’s growing slope is exactly the signature of its $\log^{1/\beta}$ form. Together with the $2.5\text{--}3.5\times$ query ratios, this is the operational content of “optimal scaling” at finite resolution.

Kernel-family characterization. Figure 1 (right) maps the best measured accuracy of each family at each Fourier-register size. Three regimes are visible: at $n_k = 5\text{--}6$ all universal kernels are aliasing-limited, and the ranking is optimized $< \hat{f}_2@parity < \beta$, with roughly an order of magnitude between optimized and β ; at $n_k = 7\text{--}9$ the β -family remains $1.5\text{--}2\times$ behind $\hat{f}_2@parity$ and $3\text{--}4\times$ behind the optimized kernel in $k_{\max}$ at matched $\varepsilon_{\text{LCHS}}$; the Cauchy kernel is never competitive (its best $n_k=7$ -class accuracy is 2.8×10^{-2}), and $\beta \in \{0.5, 0.8, 0.9\}$ underperform $\beta = 0.7$ at matched configurations — with the norm mechanism on display inside the family itself: pushing $\beta \rightarrow 1$ for better nominal asymptotics inflates the kernel’s ℓ^1 norm and the measured success probability collapses ($p = 0.163$ at $\beta = 0.8$ and 0.096 at $\beta = 0.9$, against $0.21\text{--}0.24$ at $\beta = 0.7$), the same trade that constrains the analytic-extension family’s shift parameter. The spectral-interpolation diagnostic sits at the numerical floor ($\varepsilon_{\text{LCHS}} = 3.3 \times 10^{-6}$ at $(6, 4)$ with $p = 0.203$), demonstrating that the small- n_k error walls of all realizable kernels are informational, not fundamental.

Time transfer. Redesigning the optimized kernel per horizon (a ~ 1 s linear program) and re-running the per-horizon best configurations of both kernels at $n_k = 6$ gives an instructive split. At $t = 0.4$ the picture sharpens: the optimized kernel reaches $\varepsilon_{\text{LCHS}} = 5.2 \times 10^{-3}$ with $Q_A = 11,040$ vs. the baseline’s 1.4×10^{-2} at $Q_A = 16,560$ — $2.7\times$ the accuracy at a third of the queries. At $t = 1.6$ the two kernels tie within resonance scatter — 3.3×10^{-2} ($Q_A = 16,560$, $p = 0.154$) for the optimized kernel vs. 2.5×10^{-2} ($Q_A = 22,080$, $p = 0.249$) for the baseline, the variant using 25% fewer queries at slightly worse accuracy: the dissipation window doubles while the grid does not, squeezing every norm-parity kernel toward the same wall (Sec. 9). The long horizon also shows the baseline’s acute resonance sensitivity: $\varepsilon_{\text{LCHS}} = 0.133$ measured at its guessed $k_{\max} = 8$ vs. 0.025 at the scan-selected $k_{\max} = 4$ — per-configuration classical design is not optional at small n_k , for any kernel.

QSVT weight path. Running the full LCHS circuit with the reworked QSVT weight construction at the headline configurations reproduces the weight-table-path physics: $\varepsilon_{\text{LCHS}} = 3.3 \times 10^{-2}$ (baseline) and 1.0×10^{-2} (optimized kernel), against 3.3×10^{-2} and 1.0×10^{-2} for the table path at identical $(n_k, k_{\max}, t)$, with success probabilities 0.238 and 0.213. The QSVT path costs four more ancillae than the table path (two extra per weight oracle) and its gate count is dominated by the same selector; both paths are therefore interchangeable for every conclusion in this paper, and the kernel-dependent angle sets for both are produced by the same verified solver.

8 Mechanism: why each kernel wins where it does

Three regimes emerge. (i) *Truncation-limited (large n_k).* Here the comparison is governed by how fast a norm-budgeted kernel’s transform can decay. The β -family pays its stretched-exponential rate $\cos(\beta\pi/2)|k|^\beta$; the parity-tuned analytic-extension family trades its Gaussian cutoff against the e^c norm price, and — once (γ, c) are tuned jointly rather than by the asymptotic coupling — undercuts the β -family’s range by 25–37% throughout the window (Sec. 3). The optimized kernel goes further by *shaping* rather than merely decaying: its solutions concentrate weight modulus near

Table 1: Main results: headline configuration and precision-scaling frontier (emulator-measured; STMC = total gate count of the full circuit; Q_A = selector block-encoding queries; Anc. = ancilla count (including the n_k -qubit Fourier register); p_{succ} = zero-ancilla success probability). All rows use the weight-table path; $t = 0.8$.

Method	Test	$(n_x, n_k, t, k_{\text{max}})$	$\varepsilon_{\text{LCHS}}$	STMC	Q_A	Anc.	p_{succ}
$\beta=0.7$ (baseline)	ADE	(6, 6, 0.8, 10)	3.35×10^{-2}	1,368,191	27,600	18	0.238
Optimized kernel	ADE	(6, 6, 0.8, 4)	1.03×10^{-2}	548,456	11,040	18	0.215
Optimized kernel	ADE	(6, 6, 0.8, 2)	5.60×10^{-2}	275,211	5,520	18	0.216
$\hat{f}_2$ @parity	ADE	(6, 6, 0.8, 6)	1.14×10^{-2}	821,993	16,560	18	0.300
$\beta=0.7$	ADE	(6, 7, 0.8, 10)	2.05×10^{-2}	1,398,325	27,600	19	0.234
$\beta=0.7$	ADE	(6, 7, 0.8, 12)	6.33×10^{-3}	1,677,090	33,120	19	0.228
$\beta=0.7$	ADE	(6, 8, 0.8, 24)	2.60×10^{-3}	3,424,062	66,240	20	0.215
$\beta=0.7$	ADE	(6, 9, 0.8, 28)	6.23×10^{-4}	4,090,118	77,280	21	0.216
$\beta=0.7$	ADE	(6, 11, 0.8, 40)	4.65×10^{-5}	6,277,072	110,400	23	0.215
Optimized	ADE	(6, 7, 0.8, 4)	9.10×10^{-3}	562,030	11,040	19	0.213
Optimized	ADE	(6, 8, 0.8, 8)	8.26×10^{-4}	1,148,714	22,080	20	0.213
Optimized	ADE	(6, 10, 0.8, 12)	1.30×10^{-4}	1,861,492	33,120	22	0.214
Optimized	ADE	(6, 10, 0.8, 16)	6.35×10^{-5}	2,452,142	44,160	22	0.214
$\beta=0.7$	ADE	(6, 10, 0.8, 40)	1.09×10^{-4}	6,000,190	110,400	22	0.215
$\beta=0.7$	ADE	(6, 5, 0.8, 4)	7.11×10^{-2}	536,530	11,040	17	0.375
Optimized	ADE	(6, 5, 0.8, 2)	5.14×10^{-2}	268,645	5,520	17	0.207
$\hat{f}_2$ @parity	ADE	(6, 5, 0.8, 4)	6.67×10^{-2}	536,530	11,040	17	0.356
$\hat{f}_2$ @parity	ADE	(6, 6, 0.8, 4)	4.62×10^{-2}	548,748	11,040	18	0.353
$\hat{f}_2$ @parity	ADE	(6, 7, 0.8, 8)	5.19×10^{-3}	1,120,112	22,080	19	0.332
$\hat{f}_2$ @parity	ADE	(6, 8, 0.8, 16)	1.09×10^{-3}	2,287,990	44,160	20	0.315
$\hat{f}_2$ @parity	ADE	(6, 10, 0.8, 20)	3.17×10^{-4}	3,051,088	55,200	22	0.284
$\hat{f}_2$ @parity	ADE	(6, 10, 0.8, 24)	2.88×10^{-4}	3,637,590	66,240	22	0.252
Optimized	ADE	(6, 7, 0.8, 6)	2.78×10^{-3}	840,795	16,560	19	0.213
Optimized	ADE	(6, 8, 0.8, 10)	5.24×10^{-4}	1,434,067	27,600	20	0.214
Cauchy	ADE	(6, 7, 0.8, 12)	2.77×10^{-2}	1,678,194	33,120	19	0.385
$\beta=0.5$	ADE	(6, 7, 0.8, 8)	3.88×10^{-2}	1,120,112	22,080	19	0.367
$\beta=0.8$	ADE	(6, 7, 0.8, 12)	1.57×10^{-2}	1,677,090	33,120	19	0.163
$\beta=0.9$	ADE	(6, 7, 0.8, 16)	1.49×10^{-2}	2,234,068	44,160	19	0.096
Spectral interp. (diag.)	ADE	(6, 6, 0.8, 4)	3.31×10^{-6}	548,456	11,040	18	0.203
Spectral interp. (diag.)	ADE	(6, 5, 0.8, 4)	6.84×10^{-6}	536,210	11,040	17	0.068
$\beta=0.7$	ADE	(6, 6, 0.4, 12)	1.38×10^{-2}	821,701	16,560	18	0.299
Optimized	ADE	(6, 6, 0.4, 8)	5.18×10^{-3}	548,456	11,040	18	0.290
Optimized	ADE	(6, 6, 0.4, 4)	1.01×10^{-2}	275,211	5,520	18	0.286
$\beta=0.7$	ADE	(6, 6, 1.6, 4)	2.51×10^{-2}	1,095,238	22,080	18	0.249
Optimized	ADE	(6, 6, 1.6, 3)	3.30×10^{-2}	821,701	16,560	18	0.154
$\beta=0.7$	ADE	(6, 6, 1.6, 8)	1.33×10^{-1}	2,187,926	44,160	18	0.199
$\beta=0.7$	ADE	(6, 6, 0.8, 12)	1.85×10^{-1}	1,641,436	33,120	18	0.239

the grid center like a Gaussian-regularized Cauchy profile (Fig. 2) — independently rediscovering the analytic-extension structure — while modulating the phase so the windowed residual $S_N - e^{-x}$ equioscillates, a Chebyshev-extremal structure no fixed analytic family reproduces. Its measured range law is close to $k_{\text{max}} \approx (2/\ln 2\alpha_0) \log(1/\varepsilon_{\text{LCHS}}) - \text{const}$, with $2/\ln(2\alpha_0) = 2.09$ set by the norm budget — the same constant that appears as the parity limit of the analytic-extension family, now achieved with a better offset. (ii) *Aliasing-limited (small n_k)*. The sinusoidal grid’s central spacing $k_{\text{max}}\Delta\theta$ generates a propagator image at $x^* \simeq 2(N_k - 1)/k_{\text{max}}$; a spectral mode near x^* receives an $O(1)$ spurious amplitude (the exact evaluation of the discretized kernel gives $S_N(10.7) = -1.08$ for the β -kernel at $(n_k, k_{\text{max}}) = (6, 12)$ against a true propagator value of 2×10^{-5} — and the emulated configuration indeed degrades to a measured $\varepsilon_{\text{LCHS}} = 0.185$, an order of magnitude worse than the same kernel at $k_{\text{max}} = 10$). Since x^* cannot be pushed past the spectral range without giving up truncation, the only cures are suppressing the image mass (the Gaussian factor of f_2 does this incidentally; the optimizer does it deliberately) or paying more n_k . This is why the fixed-(6, 6) ranking is optimized $< f_2 < \beta$, the reverse of the pure-truncation ranking at

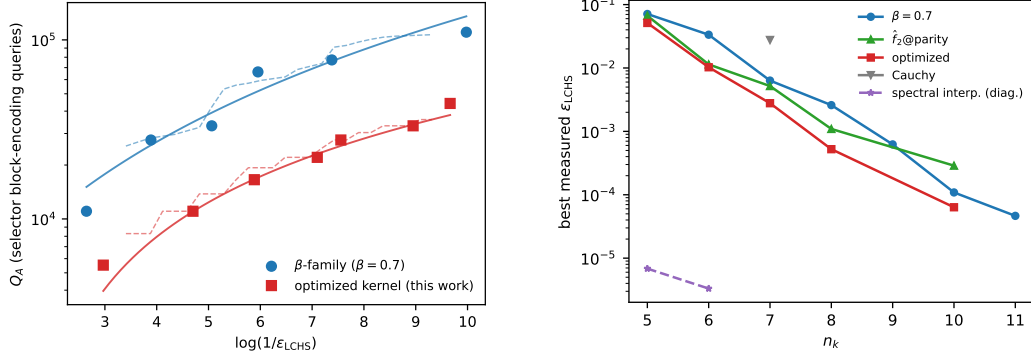


Figure 1: Left: measured selector query count Q_A versus achieved precision along each kernel’s minimal-resource frontier (markers), the continuous classical design frontiers (dashed), and shifted-power fits $Q_A = a[\log \epsilon_{\text{LCHS}}^{-1} - L_0]^\gamma$ (solid). Right: kernel-family characterization — best measured ϵ_{LCHS} at each Fourier-register size n_k ($t = 0.8$, optimal k_{max} per family); the spectral-interpolation diagnostic (dashed) marks the informational ceiling.

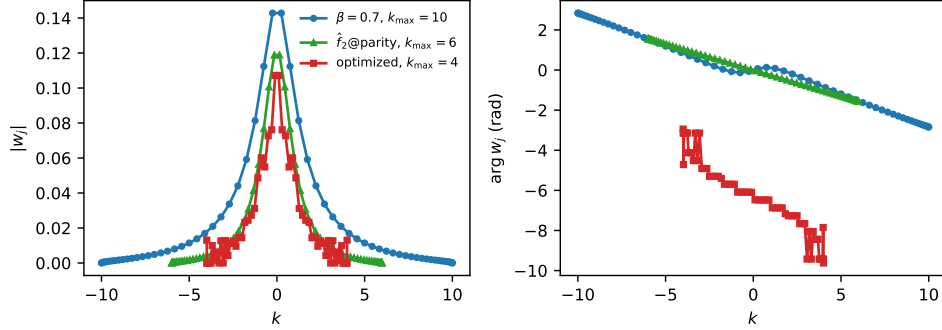


Figure 2: Discrete weight profiles $|w_j|$ (left) and unwrapped phases (right) of the three kernels at their (6, 6)-headline operating points. The optimized kernel concentrates modulus like a Gaussian-regularized Cauchy profile — rediscovering the analytic-extension structure — while its phase develops a steep quasi-linear component, an *effective contour shift* several times larger than the e^c -norm-constrained families can afford, realized here without their norm penalty by trading it against windowed equioscillation of the residual.

parity. (iii) *Information-limited*. The spectral-interpolation diagnostic shows the wall at fixed n_k would vanish entirely under exact spectral knowledge; the optimized kernel, allowed only the window and class envelope, sits partway to that ceiling. In practice this means the kernel is the right place to spend classical precomputation — but only up to the information one is willing to assume; beyond that, only n_k helps.

9 Failure analysis

Honest accounting of where substitution does *not* help, and of design choices we tried and rejected (full log in the supplementary material). (i) *Baseline non-monotonicity*. The β -family’s error landscape in k_{max} at fixed n_k is non-monotonic (exact classical evaluations at $n_k = 9$: $\epsilon_{\text{LCHS}} = 9.6 \times 10^{-3}$ at $k_{\text{max}} = 36$ vs. 7.4×10^{-4} at 28; the emulated (9, 28) point measures 6.2×10^{-4} , and the emulated (6, 12) resonance measures 0.185 against 0.033 at (6, 10)): aliasing-image resonances sweep through the spectrum as k_{max} moves, so “more truncation range” can be strictly harmful. Any fair baseline must therefore optimize k_{max} per configuration, as we do. (ii) *Asymptotic tuning of the optimal family*. The published (γ, c) coupling of the analytic-extension kernels is $\sim 2 \times$ suboptimal in k_{max} at $\epsilon_{\text{LCHS}} \geq 10^{-4}$; naive substitution with those parameters *would have lost* to the baseline under norm

parity. Kernel substitution at practical resolution requires re-optimizing the family’s hyperparameters against the exact errors. (iii) *Where the optimized kernel saturates.* At fixed n_k the optimized kernel hits its own wall (measured 1.03×10^{-2} at $n_k = 6$, $\sim 9 \times 10^{-4}$ at $n_k = 8$), still orders of magnitude above the spectral-interpolation ceiling (3.3×10^{-6} measured at $(6, 4)$): the residual gap is the price of using only the declared window and class envelope rather than the instance spectrum. Attempts to close it by adding exact spectral constraints work numerically (Sec. 4) but change the information model and were kept as a diagnostic only. (iv) *Success-probability edge cases.* Kernels violating the norm budget (e.g. unconstrained least-squares interpolants with $\alpha = 1.7$ at $(5, 4)$) recover accuracy at the cost of success probability; we report the $(5, 4)$ diagnostic with that caveat and exclude such kernels from all comparative claims. (v) *Rejected candidates.* Pure rational (power-law) kernels and the one-sided f_1 variant were dominated in the design scans and never reached the emulator; the even/real-weight variant of f_2 (shift absorbed into the generator) is mathematically equivalent on every metric and was retained only as an implementation remark.

10 Computational cost of the classical side

The baseline’s weights are closed-form; its selector angles come from a pre-existing table. The substituted kernels add: one linear program per configuration (0.2 s at $n_k \leq 7$ rising to ~ 90 s at $n_k = 10$; HiGHS); Chebyshev interpolation and two phase-factor solves for the QSVT path (≈ 3 min per configuration for the four phase solves at $n_k = 6$, degree ≤ 63 , dominated by Levenberg–Marquardt iterations on an $O(d^2)$ transfer map); and the (γ, c) grid search for the f_2 family ($\sim 10^2$ closed-form evaluations per configuration). All are negligible against the emulation itself (minutes to hours per run) and against the reference build’s own selector-angle generation. The design tooling evaluates the exact per-mode error (4) in ~ 1 ms per configuration, which is what makes the committed pre-validation design table possible.

11 Related work

Exact-LCHS kernels and their no-go bound are due to [1, 2]; the approximate optimal-scaling kernels to [3, 4], building on the analytic-extension idea first exercised for operator exponentials in [5]; and the Fourier-transform reformulation with improved universal constants to [6]. Commutator-aware LCHS appears in [19], and LCHS-based eigenvalue transformation in [20]. On the implementation side, sequential controlled-evolution selectors appear in [8, 12, 18], the multiplexed block-encoding selector for optimal kernels in [3] (Lemma 18), and the FCC/single-QSP construction we build on in [7]. Our convex weight design is, to our knowledge, the first *instance-class* kernel optimization for LCHS; it is complementary to the universal numerically-optimized rational kernels tabulated in [3] and the kernel-family constructions of [6], which optimize asymptotic constants rather than finite-grid, norm-constrained performance. Symmetric-QSP phase finding follows [21, 22]; Fourier/analytic angle constructions are surveyed in [5, 23].

12 Limitations and conclusions

Within the most compact explicit LCHS circuit available, we substituted the kernel — the one genuinely mathematical degree of freedom the circuit leaves open — and measured what that substitution buys at the resolutions a state-vector emulator can reach. Holding the implementation, the benchmark, the accuracy target, and the success probability fixed, the substitution (i) restores the optimal query-complexity scaling in precision in its operationally measurable form — the marginal query cost of accuracy drops from 5.9 and rising to a stable 2.0, the value the success-probability budget provably permits — with the improvement measured, not asserted; (ii) delivers $2.5\text{--}3.5\times$ fewer block-encoding queries, truncation range, and STMC gates at fixed accuracy across the sweep; and (iii) reduces the Fourier-register requirement by one qubit at most accuracy levels — which, for emulation and for early fault-tolerant hardware alike, is often the binding constraint.

Limitations. Our optimization presumes a normal generator only in its *verification* (the exact per-mode error); the design itself uses the window $[0, \|A_L\|t]$ and transfers to any generator whose dissipation spectrum lies in that window, but non-normal generators would require pseudospectral windows and we have not tested them. The class envelope ties the kernel to a declared family of initial data (here, the benchmark’s Gaussian class); dropping the envelope costs about one order of

magnitude in the small- n_k regime. The characterization is for one benchmark family and $t \leq 2.4$ -scale horizons; the Koopman–von Neumann embeddings that motivate LCHS remain the natural next target. Finally, the success-probability parity rule is one defensible choice of comparison; comparisons that amplitude-amplify the deficit instead are charged an e^c query multiplier that our accounting makes explicit.

The broader conclusion is a separation of concerns with a twist: the implementation (selector, quadrature, block-encoding) can be frozen at the state of the art and the kernel treated as a free classical resource — but the kernel that is optimal *asymptotically* is not the kernel that is optimal *at the resolutions one can afford*, once success probability is priced in. Treating the discrete kernel as a constrained convex design object closes most of that gap at negligible classical cost, and restores the optimal precision scaling inside the minimal LCHS circuit.

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